

PayForge — Project Folder Structure

Multi-module microservices layout · matches the 50-day sprint roadmap

This is the real, buildable folder structure for PayForge as a Maven multi-module project. The parent module (payforge-parent) declares five child modules — one shared library (common) and four independently deployable Spring Boot services. Each service module is generated at start.spring.io and then wired into the parent via its own <parent> block in its pom.xml.

```
payforge-parent/
├── pom.xml
├── docker-compose.yml
├── README.md
├── .gitignore
├── .github/
│   ├── workflows/
│   └── ci-cd.yml
├── docs/
│   ├── pci-dss-scope.md
│   ├── service-discovery-decision.md
│   ├── security-scan.md
│   └── compliance-posture.md
├── k8s/
│   ├── namespace.yaml
│   ├── ingress.yaml
│   ├── network-policy-ledger-db.yaml
│   ├── api-gateway-deployment.yaml
│   ├── api-gateway-service.yaml
│   ├── account-service-deployment.yaml
│   ├── account-service-service.yaml
│   ├── postgres-account-statefulset.yaml
│   ├── ledger-service-deployment.yaml
│   ├── ledger-service-service.yaml
│   ├── postgres-ledger-statefulset.yaml
│   ├── notification-service-deployment.yaml
│   ├── notification-service-service.yaml
│   ├── postgres-notification-statefulset.yaml
│   └── payforge-chart/
│       ├── Chart.yaml
│       ├── values.yaml
│       └── templates/
├── common/
│   ├── pom.xml
│   └── src/main/java/com/payforge/common/
│       ├── dto/
│       │   └── PaymentEvent.java
│       ├── exception/
│       │   └── InsufficientFundsException.java
├── api-gateway/
│   ├── pom.xml
│   ├── Dockerfile
│   └── src/
│       ├── main/
│       │   └── java/com/payforge/gateway/
```

parent POM — packaging=pom, lists all 5 modules
boots every service + its own Postgres + Kafka

test → build → scan → deploy, per module (Day 40, 41)

compliance & design docs (Sprint 3)
Day 27 — SAQ type + scoping decision
Day 5 — why DNS over a registry
Day 32 — dependency/image scan results
Day 34 — what's in/out of PCI DSS scope

Sprint 4 (Day 35+), empty until then

Day 38 — segmentation enforced

Helm umbrella chart (Day 39)

shared library — no @SpringBootApplication here

Day 18 — the event schema every consumer relies on

Day 11

Day 6
<parent> → payforge-parent

```

■      ■      ■      ■      GatewayApplication.java
■      ■      ■      ■      config/
■      ■      ■      ■      RouteConfig.java          DNS-based routes (Day 6)
■      ■      ■      ■      RateLimiterConfig.java    Day 21
■      ■      ■      ■      filter/
■      ■      ■      ■      JwtValidationFilter.java  Day 20 – auth enforced once, centrally
■      ■      ■      ■      resources/
■      ■      ■      ■      application.yml
■      ■      ■      ■      test/java/com/payforge/gateway/
■      ■      ■      ■      GatewayApplicationTests.java
■
■■■■ account-service/                                Day 7
■      ■      ■      ■      pom.xml
■      ■      ■      ■      Dockerfile
■      ■      ■      ■      src/
■      ■      ■      ■      main/
■      ■      ■      ■      java/com/payforge/account/
■      ■      ■      ■      AccountApplication.java
■      ■      ■      ■      controller/
■      ■      ■      ■      AccountController.java
■      ■      ■      ■      AuthController.java      Day 20 – register/login
■      ■      ■      ■      service/
■      ■      ■      ■      AccountService.java
■      ■      ■      ■      JwtService.java          Day 20
■      ■      ■      ■      repository/
■      ■      ■      ■      AccountRepository.java
■      ■      ■      ■      UserRepository.java
■      ■      ■      ■      model/
■      ■      ■      ■      Account.java              Day 9 – id, ownerName, balance, currency
■      ■      ■      ■      User.java
■      ■      ■      ■      security/
■      ■      ■      ■      SecurityConfig.java
■      ■      ■      ■      resources/
■      ■      ■      ■      application.yml
■      ■      ■      ■      db/migration/
■      ■      ■      ■      V1__create_accounts_table.sql
■      ■      ■      ■      test/java/com/payforge/account/
■      ■      ■      ■      AccountServiceTest.java
■      ■      ■      ■      AccountRepositoryIT.java  Testcontainers-backed (Day 13)
■
■■■■ ledger-service/                                Day 8 onward – the core of the system
■      ■      ■      ■      pom.xml
■      ■      ■      ■      Dockerfile
■      ■      ■      ■      src/
■      ■      ■      ■      main/
■      ■      ■      ■      java/com/payforge/ledger/
■      ■      ■      ■      LedgerApplication.java
■      ■      ■      ■      controller/
■      ■      ■      ■      PaymentController.java    Day 10 – POST /api/v1/payment-intents
■      ■      ■      ■      RefundController.java     Day 45
■      ■      ■      ■      service/
■      ■      ■      ■      PaymentService.java       Day 10 – idempotency logic lives here
■      ■      ■      ■      FraudRuleService.java     Day 45
■      ■      ■      ■      TokenizationService.java  Day 28 – PCI DSS scoping
■      ■      ■      ■      repository/
■      ■      ■      ■      TransactionRepository.java
■      ■      ■      ■      LedgerEntryRepository.java
■      ■      ■      ■      model/
■      ■      ■      ■      Transaction.java          idempotencyKey (unique)
■      ■      ■      ■      LedgerEntry.java          double-entry: +credit / -debit
■      ■      ■      ■      PaymentMethod.java        Day 28 – token, last4, cardBrand only

```

```

■      ■      ■      ■      ■■■■ PaymentStatus.java          state machine enum (Day 23)
■      ■      ■      ■■■■ client/
■      ■      ■      ■■■■ AccountServiceClient.java Day 8 – Feign, DNS-based, resilient (Day 23)
■      ■      ■      ■■■■ kafka/
■      ■      ■      ■■■■ PaymentEventProducer.java Day 18
■      ■      ■■■■ resources/
■      ■      ■■■■ application.yml
■      ■      ■■■■ db/migration/
■      ■      ■■■■ V1__create_ledger_tables.sql
■      ■■■■ test/java/com/payforge/ledger/
■      ■■■■ PaymentServiceTest.java
■      ■■■■ IdempotencyIT.java          Day 10 – proves the double-charge fix works
■      ■■■■ SagaCompensationIT.java    Day 24 – the compensating-action test
■
■■■■ notification-service/          Day 19
■■■■ pom.xml
■■■■ Dockerfile
■■■■ src/
■■■■ main/
■      ■■■■ java/com/payforge/notification/
■      ■      ■■■■ NotificationApplication.java
■      ■      ■■■■ kafka/
■      ■      ■      ■■■■ AuditConsumer.java          Day 19 – @KafkaListener, manual ack
■      ■      ■■■■ webhook/
■      ■      ■      ■■■■ WebhookDeliveryService.java Day 22 – HMAC-signed, retry-with-backoff
■      ■      ■      ■■■■ WebhookEndpointRepository.java
■      ■      ■■■■ model/
■      ■      ■■■■ AuditLog.java          insert-only DB role (Day 31)
■      ■      ■■■■ WebhookEndpoint.java
■      ■■■■ resources/
■      ■■■■ application.yml
■      ■■■■ db/migration/
■      ■■■■ V1__create_audit_log_table.sql
■■■■ test/java/com/payforge/notification/
■■■■ AuditConsumerTest.java

```

How the pieces connect

common is a plain library jar — it has no `@SpringBootApplication` and is never run on its own. The other four modules depend on it for shared DTOs (like `PaymentEvent`) and shared exception types, so the event schema and error handling stay consistent everywhere.

api-gateway, **account-service**, **ledger-service**, **notification-service** are each independently buildable and independently deployable Spring Boot applications — that is the actual definition of "microservices" this project uses, not just a naming convention. Each has its own `pom.xml` with a `<parent>` block pointing back to `payforge-parent`, its own `Dockerfile`, and — for the three business services — its own Postgres schema, so no service can reach another's database directly.

k8s/ and **docs/** start empty and fill in during Sprints 3 and 4 — they're listed here in full so you can see the target shape from Day 4, even though most of their contents don't exist yet.